Evaluating Adaptive Region Acquisition for Spatial Transcriptomics Reconstruction

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Abstract

Region-based spatial transcriptomics measures only part of a tissue section, and several recent methods choose the next region to measure from the expression already measured. Whether this adaptivity helps depends on the goal. Reconstructing expression across the whole section and capturing regions of high molecular heterogeneity (ROI detection) reward different choices, and sensor-placement theory predicts little benefit for the first. We tested both goals in a retrospective simulator built from 22 human HEST-1k slides (7 Visium HD, 15 Visium; 18 studies; four tissues). Image encoder and imputer were held fixed, and adaptive policies were compared with strong non-adaptive placement. Twenty-four primary comparisons, with decision rules, were fixed before the run; tests resample whole studies and are Holm-adjusted. For reconstruction, adaptive policies were worse than the best non-adaptive baseline at budgets of 0.5–2% of regions (mean differences -0.128 to -0.040 PCC over the top-50 genes; adjusted $p < 0.001$) and close to it at 10–20% (-0.002). A policy that selects directly on measured values was worse at every budget tested (-0.136 to -0.165). For ROI detection, that policy raised recall over the best coverage baseline by $+0.001$ to $+0.042$, but no gain remained significant after adjustment ($p \geq 0.36$), and none reached significance under batched or contiguous acquisition. In secondary and exploratory analyses, a non-adaptive diversity sampler gained a similar amount of recall, and the gains were concentrated in the Visium HD slides. On synthetic slides with strong ROI signal the same protocol registered an advantage of $+0.10$ recall, so it can detect one when it exists. At these budgets and on these slides, non-adaptive placement matched or exceeded the adaptive policies for reconstruction, and no adaptive advantage for detection was supported.

1 Introduction

Whole-transcriptome spatial assays are costly, and region-of-interest platforms such as GeoMx DSP [9] profile only an operator-chosen subset of a section. Regions that are not profiled can be reconstructed afterwards, usually by predicting expression from co-registered hematoxylin and eosin (H&E) images [5, 11]. The choice of regions therefore determines two things: how well the rest of the section can be predicted, and whether biologically distinctive regions are measured at all.

Recent methods make this choice sequentially. SOFisher [7] trains a reinforcement-learning policy that moves fields of view toward regions enriched for a target signal, using the expression measured so far. SCR^2-ST [15] uses reinforcement learning to choose sampling locations for expression prediction, with a reward dominated by coverage of a single-cell reference manifold. S2-omics [14] takes the opposite approach: it selects ROIs from histology before any measurement in order to capture molecular heterogeneity, and its preprint argues that sequential designs that wait for measured expression are hard to run in practice. These methods pursue different objectives. SOFisher and S2-omics aim to capture particular regions; reconstruction aims to predict expression everywhere.

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The distinction matters because the two objectives lead to different expectations. For reconstruction, Gaussian-process sensor placement gives a reason to expect little from adaptivity: with a fixed kernel, the posterior variance at each location depends on where measurements were taken but not on the values observed, so the best adaptive and non-adaptive designs coincide [6]. For detection there is no such argument, because the targets are defined by the values themselves. A sequential policy may help one objective and not the other, and an evaluation that reports only one cannot tell which.

We therefore asked three questions under matched conditions, with the image encoder and imputer fixed and strong non-adaptive placement rules as comparators: **Q1**, does conditioning region choice on measured expression improve whole-section reconstruction? **Q2**, does it improve the capture of molecularly heterogeneous regions? **Q3**, does any advantage survive the batched and contiguous acquisition that real platforms impose? Because a null result is only informative if the protocol could have detected an effect, we first ran synthetic positive controls.

Under the pre-specified rules, the reconstruction null was retained for Q1, and a detection advantage was not supported for Q2 or Q3. Adaptive policies were worse than or close to non-adaptive placement for reconstruction. Their detection gains were small, did not survive correction, and were approached by a non-adaptive design. The synthetic controls show that the protocol registers a detection advantage when the ROI signal in measured expression is strong. The contribution is an evaluation, not a new method: a reproducible protocol on 22 human slides from 18 studies with analyses fixed in advance, and the finding that, at the budgets tested, no adaptive policy we evaluated had a supported advantage over strong non-adaptive placement for either objective.

2 Related work

Predicting expression from histology. Expression has been predicted from H&E by contrastive retrieval [11], flow matching over spatial neighbourhoods [4], diffusion [16], and autoregressive generation [10], benchmarked on HEST-1k [5]. We hold the predictor fixed and vary only which regions it receives.

Choosing what to measure. Gene-panel selectors fix a set of genes [2, 8]. Region selectors range from one-shot image rankers [12] and histology-based ROI selection for molecular heterogeneity [14] to sequential reinforcement-learning samplers [7, 15]. We add Gaussian-process placement baselines and score the reconstruction and detection objectives on the same slides.

Active acquisition and sensor placement. Active feature acquisition treats measurement as a sequential policy [13]. Variance- and mutual-information-greedy placement are submodular and near-optimal for reconstructing a Gaussian-process field [6], which makes them the natural non-adaptive comparators for Q1.

3 Methods

3.1 Slides and acquisition unit

We used 22 human slides from HEST-1k [5]: prostate (6), kidney (4), lung (6), and colon (6); 7 are Visium HD and 15 are standard Visium (Appendix A). Selection rules were fixed before the results were seen: human tissue, whole-transcriptome assay, at least 1,000 spots under tissue, source image at most 0.6 μm/px, one slide per specimen, and at most two slides per study. The last two rules limit dependence between slides, and the image-resolution rule keeps the image inputs comparable. The 22 slides come from 18 HEST studies, which we treat as the independent units in all tests.

The acquisition unit ("region") is one native spot. Visium spots are 55 μm in diameter on a 100 μm pitch. The Visium HD slides, as distributed by HEST, have a median spot spacing of 128 μm on our slides. We used the spot directly because it is the unit at which HEST provides both expression and an image patch, and because a fixed square grid would have pooled different numbers of spots on slides with different pixel sizes. Coordinates were converted to microns with each slide's estimated pixel size. Slides contain 1,766 to 9,080 regions. Each region's image input is the HEST 224 px

patch at 0.5 μm/px centred on the spot, embedded with frozen UNI2-h [1]. Expression is log1p-transformed, and genes detected in fewer than 10% of regions are removed.

3.2 Objectives

A policy reveals regions up to a budget, expressed as a fraction of the slide's regions (0.5% to 20%), and each revealed region exposes its measured expression. *Reconstruction* (Q1): a fixed imputer predicts the unprofiled regions from the profiled ones, and we score the per-gene Pearson correlation over unprofiled regions, averaged over the top-50 highly variable genes (PCC), the HEST-Benchmark target [5]. *ROI detection* (Q2): following the heterogeneity objective of S2-omics [14], ROIs are the 20% of regions with the highest local molecular heterogeneity, defined as the variance of top-50-gene expression across a region's 8 nearest spatial neighbours, averaged over genes; we score recall, the fraction of ROI regions that were profiled. ROIs and gene sets come from ground truth and are used only for scoring.

3.3 Imputer

The primary imputer (graphflow) maps image principal components through 512 random Fourier features, fits a ridge regression to the profiled regions' expression, and then runs 20 steps of message passing on a fixed 8-nearest-neighbour graph in image-and-space features, holding profiled regions at their measured values. We used one fixed imputer for every policy so that differences between policies reflect selection alone. One property of this imputer governs how the policies should be read: its *predictions* depend on the measured values, but its *uncertainty*, one minus the graph influence reaching a region from profiled nodes, depends only on which regions are profiled. A k-nearest-neighbour and a regression-kriging imputer were run on the earlier slide set only (Appendix D).

3.4 Policies

The policies fall into four groups according to what they test. *Non-adaptive designs*: random (5 seeds), farthest-point in space (grid), k-means on image embeddings with medoids, a reimplementation of a one-shot image ranker [12], and variance-greedy and mutual-information (MI)-greedy placement with an RBF kernel on joint image and coordinate features [6]. *Sequential, value-independent*: error-reduction selects the region whose 15 nearest unprofiled neighbours in the imputer's feature space have the largest summed imputer uncertainty, and maximum-uncertainty selects the region of highest uncertainty. Under graphflow their choices depend on the profiled set but not on the measured values, so they are fixed designs computed sequentially. *Value-conditioned*: each candidate is scored by the standardized kernel-weighted local variance of the measured expression around it plus 0.5 times its standardized distance to the nearest profiled region, starting from the region nearest the slide centroid. Its choices depend on the measured values, so it is the direct test of Q1 and Q2. *Competitor objectives*: a diversity sampler (farthest-point in joint image and space, after the coverage-dominated reward of SCR^2-ST [15]; non-adaptive) and an ROI-seeking sampler (moves toward measured regions whose expression deviates most from the measured mean, after SOFisher [7]; adaptive). These reimplement the stated objectives inside our simulator and are not the authors' trained models.

3.5 Positive controls

To check that a null result would not simply reflect an insensitive protocol, we generated synthetic 20×20 slides with image-unpredictable, spatially clustered structure (8 slides per setting, 10% budget, k-NN imputer). For reconstruction we varied the correlation length of the hidden structure and added a non-deployable *difficulty oracle* that orders regions by true local heterogeneity weighted by distance to regions already chosen. It uses ground truth but is a heuristic, not a proven upper bound. For detection we varied the amplitude α with which the ROIs appear in measured expression. These controls use slide-level bootstrap intervals over the 8 synthetic slides and are descriptive.

3.6 Comparators and statistical analysis

For reconstruction, a policy's comparator on each slide is the best of the five non-adaptive policies other than random ("best non-adaptive"); for detection and Q3 it is the best of k-means, grid, and MI-

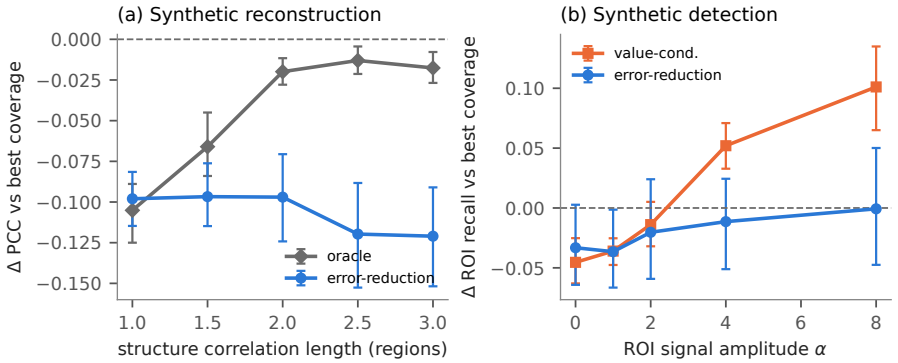


Figure 1: Synthetic controls (8 slides per point, 10% budget, mean and 95% bootstrap CI over slides). (a) Reconstruction margin over the best coverage baseline versus the correlation length of the hidden structure, for a ground-truth difficulty oracle and the error-reduction policy. (b) Detection-recall margin versus ROI signal amplitude. The dashed line marks zero.

greedy ("best coverage"). Taking the per-slide maximum favours the baseline, so a positive margin is conservative evidence for adaptivity. For each margin (policy minus comparator, per slide) we report the mean, a 95% percentile bootstrap CI that resamples whole studies (5,000 resamples), and an exact two-sided sign-flip test that flips the signs of whole studies (all 2^{18} patterns). Resampling studies rather than slides accounts for slides from the same study being related.

The analysis plan was committed to the repository before the 22-slide run. Results fall into four categories, which we keep distinct throughout. *Confirmatory*: a primary family of 24 margins: error-reduction versus best non-adaptive at six budgets; value-conditioned versus best coverage for reconstruction and for detection at four budgets; and the value-conditioned reconstruction and detection margins in the five acquisition settings of Q3. P-values in this family are Holm-adjusted [3] (p_H). The pre-specified rules were: the reconstruction null is retained if the primary reconstruction margins are at most zero and none is significantly positive after Holm adjustment; a detection advantage is claimed only for a positive margin with $p_H < 0.05$. *Secondary*: other margins stored by the same experiments (competitor samplers, gene-set sensitivity), Holm-adjusted within their experiment where stated. *Descriptive*: per-tissue means, the synthetic controls, the five-slide imputer comparison, and the native SOFisher run. *Exploratory or post hoc*: the platform split, the maximum-uncertainty margin over the best non-adaptive baseline, and the equivalence tolerance of 0.01 PCC, all defined after the results were seen.

4 Results

4.1 The protocol can register an adaptivity advantage

The synthetic controls bear on how to read the real-slide results. For reconstruction, even the difficulty oracle stayed below the best coverage baseline at every correlation length: its margin was -0.105 and -0.066 PCC at the two shortest lengths and between -0.013 and -0.020 at the three longer ones, with every CI below zero (Figure 1a). For detection, the value-conditioned margin over best coverage grew with the ROI amplitude, from -0.045 at $\alpha = 0$ to $+0.052$ (95% CI $+0.033$ to $+0.071$) at $\alpha = 4$ and $+0.101$ ($+0.065$ to $+0.135$) at $\alpha = 8$, and was positive on all 8 slides in both of the latter settings (Figure 1b). The error-reduction policy, which targets the imputer's uncertainty (under k-NN, the variance of measured expression among neighbours) rather than heterogeneity, stayed at or below zero.

Two consequences follow. The simulator and recall metric can show a value-adaptivity advantage for detection, so a null result on real slides is informative about the slides rather than only about the protocol. And in this synthetic setting, even an ordering informed by true local difficulty did not improve reconstruction, which anticipates the answer to Q1.

Table 1: Absolute reconstruction accuracy (PCC over the top-50 genes on unprofiled regions), mean over 22 slides, graphflow imputer. "Best per slide" is the highest-scoring non-adaptive policy on each slide. Value-conditioned selection was run at 2–20% only.

	Non-adaptive			Adaptive		
Budget	best per slide	MI-greedy	k-means	error-red.	max-unc.	value-cond.
0.5%	0.470	0.412	0.449	0.343	0.348	–
1%	0.550	0.522	0.534	0.467	0.462	–
2%	0.607	0.599	0.598	0.567	0.562	0.471
5%	0.660	0.659	0.653	0.650	0.648	0.513
10%	0.687	0.685	0.679	0.684	0.685	0.535
20%	0.709	0.707	0.696	0.708	0.710	0.544

Table 2: Primary (confirmatory) margins over 22 slides (18 study clusters). Δ is the mean per-slide difference with a 95% bootstrap CI that resamples studies; p_H is the exact study-level sign-flip p-value, Holm-adjusted over the 24 pre-specified primary margins (this table and Table 4).

	Reconstruction (PCC, top-50 genes)				Detection (ROI recall)	
	error-reduction − best non-adapt.		value-cond. − best coverage		value-cond. − best coverage	
Budget	Δ [95% CI]	p_H	Δ [95% CI]	p_H	Δ [95% CI]	p_H
0.5%	-0.128 $[-0.162, -0.096]$	<0.001	–	–	–	–
1%	-0.083 $[-0.105, -0.064]$	<0.001	–	–	–	–
2%	-0.040 $[-0.054, -0.027]$	<0.001	-0.136 $[-0.171, -0.106]$	<0.001	$+0.001$ $[-0.004, +0.006]$	0.723
5%	-0.011 $[-0.018, -0.004]$	0.057	-0.148 $[-0.180, -0.119]$	<0.001	$+0.013$ $[+0.002, +0.024]$	0.362
10%	-0.002 $[-0.007, +0.001]$	0.687	-0.152 $[-0.186, -0.122]$	<0.001	$+0.021$ $[+0.003, +0.037]$	0.362
20%	-0.002 $[-0.004, +0.001]$	0.687	-0.165 $[-0.217, -0.125]$	<0.001	$+0.042$ $[+0.006, +0.077]$	0.362

4.2 Q1: Reconstruction

No sequential policy improved reconstruction at any budget. Absolute accuracy is in Table 1, and the confirmatory margins are in Table 2 and Figure 2a. The error-reduction policy was below the best non-adaptive baseline by 0.128, 0.083, and 0.040 PCC at 0.5%, 1%, and 2% ($p_H < 0.001$; lower in all 18 studies at 0.5% and 1% and in 17 of 18 at 2%) and by 0.011 at 5% ($p_H = 0.057$). At 10% and 20% the margins were -0.002. No primary reconstruction margin was positive, so the reconstruction null is retained under the pre-specified rule.

The imputer property noted in Methods frames this result. Under graphflow the error-reduction and maximum-uncertainty policies are fixed designs computed one region at a time, so Q1 here compares one kind of fixed design with others; at small budgets the uncertainty-driven designs reconstructed worse than k-means or MI-greedy placement, and the gap closed as the budget grew. In a post hoc check, the 90% study-cluster CIs at 10% and 20%, $[-0.006, +0.001]$ and $[-0.003, +0.001]$, lie within ± 0.01 PCC, so at that tolerance (chosen after seeing the data) the error-reduction policy and the best non-adaptive baseline are equivalent. The maximum-uncertainty policy behaved similarly (Table 1); in a post hoc comparison it was $+0.001$ above the best non-adaptive baseline at 20% (95% CI -0.001 to $+0.003$, unadjusted $p = 0.35$).

The value-conditioned policy, the direct test of value-adaptivity, was below best coverage at every budget, by 0.136 to 0.165 PCC ($p_H < 0.001$). It reached 0.544 at 20% (Table 1), against 0.696 to 0.707 for the individual coverage policies. A plausible reading, which we did not test directly, is that steering measurements toward locally variable areas reduced the spread that reconstruction relies on.

4.3 Q2: ROI detection

The value-conditioned policy raised recall over best coverage by $+0.001$, $+0.013$, $+0.021$, and $+0.042$ at 2%, 5%, 10%, and 20% (Table 2, Figure 2b). The unadjusted p-values at 5–20% were 0.040 to 0.044; after Holm adjustment over the primary family, all were 0.36. Under the pre-specified rule, these margins do not establish a detection advantage. To put the estimates in scale, the coverage policies captured ROIs at roughly the rate implied by the budget (recall 0.200–0.232

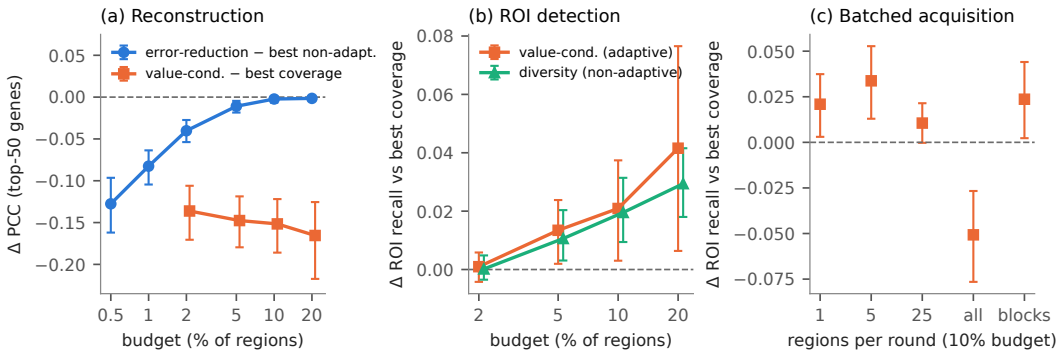


Figure 2: Margins over 22 slides, mean and 95% CI resampling studies. (a) Reconstruction (Q1). (b) ROI-detection recall (Q2) for the value-conditioned policy and the non-adaptive diversity sampler. (c) Detection margin of the value-conditioned policy at 10% budget as acquisition is batched (regions per round; "all" is a single round) or made contiguous ("blocks": each round reveals the 7 regions nearest a chosen seed) (Q3).

Table 3: Absolute ROI-detection recall (fraction of the top-20% most heterogeneous regions profiled), mean over 22 slides. †Reimplementations of the stated objectives of SCR^2-ST (diversity, non-adaptive) and SOFisher (ROI-seeking, adaptive) in our simulator. *Profiles regions in order of true heterogeneity; it uses ground truth and bounds recall from above.

	Non-adaptive coverage			Competitor objectives			
Budget	k-means	grid	MI-greedy	diversity†	ROI-seeking†	value-cond.	oracle*
2%	0.018	0.022	0.018	0.026	0.021	0.027	0.100
5%	0.050	0.051	0.046	0.067	0.050	0.070	0.250
10%	0.110	0.103	0.094	0.136	0.104	0.137	0.500
20%	0.232	0.204	0.200	0.266	0.232	0.278	0.999

at 20%), the value-conditioned policy reached 0.278, and the ground-truth ordering reached 0.999 (Table 3).

Two further analyses bear on whether the small positive estimates should be attributed to adaptivity. In the secondary competitor analysis, the non-adaptive diversity sampler gained a similar amount over best coverage: $+0.020$ (95% CI $+0.009$ to $+0.031$) at 10% and $+0.029$ ($+0.018$ to $+0.042$) at 20%, with Holm-adjusted p of 0.015 and 0.002 within that experiment, where the value-conditioned margins had adjusted $p = 0.36$. This suggests that much of the resolvable gain does not require adaptivity. In an exploratory platform split (Appendix B), the value-conditioned detection margins at 10% and 20% were $+0.048$ and $+0.099$ on the Visium HD slides (7 slides, 5 studies; with 5 clusters the smallest attainable p is 0.0625) and $+0.008$ and $+0.015$ on Visium (15 slides, 13 studies; $p = 0.43$ and 0.47). The HD subset is too small to test, and we cannot tell whether the difference reflects the platform or the particular studies.

The ROI-seeking sampler, our reimplementation of SOFisher's objective, did not improve recall (-0.012 to -0.005, all CIs including zero) and lowered reconstruction PCC by 0.19 to 0.28. This says little about SOFisher itself: run natively from the authors' code on its bundled mouse cortex data for 5 million steps, its evaluation reward for captured target regions rose from a median of -238 early in training to $+147$ late (descriptive; Appendix E). The native method learns target capture on its own task, so the port's result on our slides should not be read as a test of SOFisher. We could not run SCR^2-ST natively because its processed training data are not publicly available.

4.4 Q3: Batched and contiguous acquisition

Q2 assumed that a policy sees each measurement before choosing the next region, which real platforms do not allow. At 10% budget we therefore revealed regions in rounds of 1, 5, or 25, or in a single round, with the policy using only measurements available at the start of each round,

Table 4: Value-conditioned policy minus best coverage at 10% budget under batched and contiguous acquisition (22 slides; Δ with 95% CI resampling studies; p_H Holm-adjusted over the primary family).

	Reconstruction (PCC)		Detection (ROI recall)	
Acquisition setting	Δ [95% CI]	p_H	Δ [95% CI]	p_H
Sequential (1 region per round)	-0.152 $[-0.186, -0.122]$	<0.001	$+0.021$ $[+0.003, +0.037]$	0.362
Rounds of 5 regions	-0.129 $[-0.175, -0.094]$	<0.001	$+0.034$ $[+0.013, +0.053]$	0.073
Rounds of 25 regions	-0.130 $[-0.176, -0.095]$	<0.001	$+0.011$ $[-0.000, +0.021]$	0.362
Single round (whole budget)	-0.236 $[-0.288, -0.188]$	<0.001	-0.051 $[-0.077, -0.027]$	0.008
Contiguous blocks of 7 regions	-0.087 $[-0.116, -0.063]$	<0.001	$+0.024$ $[+0.002, +0.044]$	0.362

and separately in contiguous blocks of 7 regions around a policy-chosen seed (Table 4, Figure 2c). Reconstruction margins were negative in every setting ($p_H < 0.001$). Detection margins were $+0.021$, $+0.034$, and $+0.011$ for rounds of 1, 5, and 25 and $+0.024$ for contiguous blocks, none with $p_H < 0.05$ (rounds of 5: $p_H = 0.073$). With a single round the policy has no measurements and falls back to its starting rule, taking the regions nearest the slide centroid; this compact block captured fewer ROIs than the coverage policies (-0.051, $p_H = 0.008$). No acquisition setting produced a supported detection advantage, so Q3 does not change the answer to Q2.

4.5 Sensitivity of the Q1 and Q2 results

Gene set (secondary). Choosing the top-50 genes from the full slide uses information a prospective experiment would lack. Choosing them from a random 15% reference subset changed the value-conditioned reconstruction margins by at most 0.003, and scoring all filtered genes gave smaller but still negative margins (-0.033 to -0.079, all CIs below zero; Appendix C). The Q1 result does not depend on the gene choice. *Tissue* (descriptive). The value-conditioned reconstruction margins were negative in all four tissues at every budget, as were the error-reduction margins at 0.5–5%; at 10–20% the error-reduction margin in kidney was $+0.001$ and $+0.000$. The detection margins were positive in kidney, lung, and colon and between -0.006 and -0.001 in prostate at 2–10% (Appendix B).

5 Discussion

What the results allow us to conclude. For reconstruction, the evidence supports a specific statement. On these slides and at these budgets, no policy we tested reconstructed better than the best non-adaptive design, and the policy that used measured values most directly reconstructed worst. This agrees with the sensor-placement argument but does not test it. Our sequential policies under graphflow were value-independent by construction, and the value-conditioned policy optimized a different quantity (local variance) rather than the posterior error. The positive control adds that even an ordering informed by true local difficulty did not help. Together these observations suggest that, for whole-section reconstruction at these budgets, spreading measurements evenly mattered more than steering them by what had been measured.

For detection, the results are weaker in both directions. The synthetic control shows that value-adaptivity can improve detection when ROIs announce themselves strongly in measured expression. On the real slides, the estimated gain was at most 0.042 recall at 20% budget, against a gap of about 0.72 to the ground-truth ordering. It did not survive correction, did not appear under any realistic acquisition setting, and was approached by a non-adaptive diversity design. We therefore cannot rule out a small detection benefit, but we found no evidence that adaptivity, as opposed to a better spread of measurements, produced one.

Practical implication. For budgets of 0.5–20% of spots on slides like these, a placement rule computed before any measurement was a sufficient comparator. MI-greedy or k-means placement was as accurate for reconstruction as any adaptive policy we tested, and a non-adaptive diversity design reached recall within 0.012 of the value-conditioned policy (0.266 against 0.278 at 20% budget) without waiting for measurements between rounds. These results suggest that adaptive acquisition methods be compared with such baselines, on the objective they are meant to serve.

Limitations. The simulator is retrospective: a revealed region returns its measured expression without added noise, batch effects, or turnaround time, conditions that, if anything, favour adaptive policies. The acquisition unit is a spot or HD pseudo-spot, which may be smaller than the ROIs used on platforms such as GeoMx, and the HD and Visium units differ in spacing. The 22-slide results use one imputer; the k-NN and kriging comparisons rest on 5 slides. The ROI definition is one of several reasonable choices, and ROIs from expert annotation could behave differently. The competitor samplers were evaluated as reimplementations of their objectives, not as the authors' trained systems; SOFisher was run natively only on its own data, and SCR^2-ST not at all. The equivalence tolerance and the platform split were chosen after the results were seen. Budgets above 20% were not tested.

6 Conclusion

Q1: across 22 human slides from 18 studies, conditioning region choice on measured expression did not improve whole-section reconstruction. Adaptive policies were worse than the best non-adaptive placement at 0.5–2% budgets, within 0.01 PCC of it at 10–20% (post hoc tolerance), and the value-conditioned policy was worse at every budget tested. Q2: a detection advantage was not supported; recall gains of up to 0.042 did not survive correction for 24 comparisons, and a non-adaptive diversity sampler gained a similar amount. Q3: no batched or contiguous acquisition setting produced a supported detection advantage. The synthetic controls show that the protocol can register a detection advantage when ROI signal in measured expression is strong, so these nulls describe the tested slides and budgets rather than an insensitive test. In practice, at these scales, non-adaptive placement matched or exceeded the adaptive policies we evaluated for reconstruction, and none of those policies had a supported detection advantage over it.

Reproducibility. Code, configuration, result files, the pre-registered analysis plan, and the scripts that regenerate every table and figure (`scripts/14_power_summary.py`, `scripts/15_paper_assets.py`) are available at https://github.com/ayushdeo/STARE.

A Slides

Table 5 lists the slides. "Cluster" indexes the HEST study used as the resampling unit; "Pitch" is the median nearest-neighbour spacing of regions in microns.

Table 5: The 22 slides (HEST-1k v1.3 metadata).

Tissue	HEST ID	Platform	Spots	μm/px	Pitch (μm)	Cluster	Study (HEST dataset title)
prostate	TENX174	Visium HD	2385	0.274	128	1	Visium HD Spatial Gene Expression Library, Human Prostate ...
prostate	TENX40	Visium	4371	0.345	100	2	Human Prostate Cancer, Adenocarcinoma with Invasive Carcin...
prostate	MEND160	Visium	4079	0.573	100	3	Spatially resolved clonal copy number alterations in benig...
prostate	TENX46	Visium	3043	0.344	100	4	Human Prostate Cancer, Acinar Cell Carcinoma (FFPE)
prostate	TENX41	Visium	2543	0.345	100	5	Normal Human Prostate (FFPE)
prostate	MEND144	Visium	4074	0.341	100	3	Spatially resolved clonal copy number alterations in benig...
kidney	TENX173	Visium HD	2684	0.274	128	6	Visium HD Spatial Gene Expression Library, Human Kidney (F...
kidney	TENX71	Visium	5936	0.274	100	7	Human Kidney, 11 mm Capture Area (FFPE)
kidney	INT21	Visium	4975	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
kidney	INT19	Visium	4948	0.457	100	8	Tertiary lymphoid structures generate and propagate anti-t...
lung	TENX171	Visium HD	1766	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX170	Visium HD	1877	0.274	128	9	Visium HD Spatial Gene Expression Libraries, Post-Xenium, ...
lung	TENX163	Visium HD	2487	0.274	128	10	Visium HD Spatial Gene Expression Library, Human Lung Canc...
lung	TENX72	Visium	6195	0.274	100	11	Human Lung Cancer, 11 mm Capture Area (FFPE)
lung	TENX62	Visium	3858	0.273	100	12	Human Lung Cancer (FFPE)
lung	MISC20	Visium	3316	0.438	100	13	A spatially resolved atlas of the human lung characterizes...
colon	TENX156	Visium HD	2179	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX155	Visium HD	2281	0.274	128	14	Characterization of immune cell populations in the tumor m...
colon	TENX70	Visium	9080	0.274	100	15	Human Colorectal Cancer, 11 mm Capture Area (FFPE)
colon	TENX92	Visium	6518	0.274	100	16	Visium CytAssist Gene Expression Libraries of Post-Xenium ...
colon	MISC47	Visium	4818	0.276	100	17	COLON MAP: Colon Molecular Atlas Project
colon	ZEN48	Visium	2385	0.459	100	18	Charting the Heterogeneity of Colorectal Cancer Consensus ...

Changes from the earlier seven-slide analysis. An earlier version of this analysis used seven Visium HD slides and described regions as 112 μm squares. Two corrections apply. First, the current HEST metadata lists two of those slides as non-human (TENX186, *Mus musculus*; TENX175, mixed human and mouse); both are excluded here. Second, the loader had assumed 0.5 μm/px, whereas the HD images are about 0.274 μm/px, so the regions were about 61 μm squares containing exactly

one pseudo-spot each. The present analysis defines the unit as the spot directly. On the five retained slides, the per-slide value-conditioned margins (reconstruction and detection, four budgets) differ between the two runs by a median of 0.015 (maximum 0.096), from the use of true spot positions instead of grid centres; 3 of 40 change sign, two of them smaller than 0.005 in magnitude and one detection margin moving from -0.050 to $+0.015$. With seven slides the smallest attainable exact p-value is 0.016, so none of the earlier margins could have survived a correction for 24 comparisons.

B Platform and tissue splits

The platform split is exploratory (not in the analysis plan); the per-tissue means are descriptive.

Table 6: Selected margins by platform (with unadjusted study-level sign-flip p) and by tissue (mean). With 5 HD studies the smallest attainable p is 0.0625.

	HD (n=7, 5 cl.)	Visium (n=15, 13 cl.)	prostate (n=6)	kidney (n=4)	lung (n=6)	colon (n=6)
Recon., error-red. (1%)	-0.113 (0.062)	-0.068 (<0.001)	-0.078	-0.085	-0.081	-0.088
Recon., value-cond. (10%)	-0.103 (0.062)	-0.174 (<0.001)	-0.115	-0.171	-0.141	-0.185
Detect., value-cond. (10%)	$+0.048$ (0.062)	$+0.008$ (0.426)	-0.001	$+0.022$	$+0.036$	$+0.027$
Detect., value-cond. (20%)	$+0.099$ (0.062)	$+0.015$ (0.469)	$+0.015$	$+0.024$	$+0.052$	$+0.070$
Detect., rounds of 5 (10%)	$+0.049$ (0.250)	$+0.026$ (0.044)	$+0.016$	$+0.012$	$+0.046$	$+0.053$

C Gene-set sensitivity

Table 7: Value-conditioned minus best coverage reconstruction margin (PCC) under three gene sets, 22 slides, 95% CI resampling studies (secondary).

Budget	per-slide top-50 HVG	reference-subset top-50 HVG	all filtered genes
2%	-0.136 $[-0.171, -0.106]$	-0.134 $[-0.168, -0.103]$	-0.032 $[-0.047, -0.021]$
5%	-0.148 $[-0.180, -0.119]$	-0.148 $[-0.182, -0.118]$	-0.052 $[-0.068, -0.040]$
10%	-0.152 $[-0.186, -0.122]$	-0.153 $[-0.189, -0.122]$	-0.068 $[-0.087, -0.052]$
20%	-0.165 $[-0.217, -0.125]$	-0.166 $[-0.217, -0.126]$	-0.079 $[-0.101, -0.061]$

D Imputer comparison on the earlier slide set

The k-NN and kriging imputers were run only on the earlier slide set. Table 8 reports the error-reduction policy minus the best non-adaptive baseline on the five human slides from that run (legacy unit, which equals one pseudo-spot per region on these slides), with the number of slides on which the margin was positive. With five slides these values are descriptive.

Table 8: Error-reduction minus best non-adaptive (PCC), mean over 5 human Visium HD slides (positive/total).

Budget	kNN	kriging	graphflow
0.5%	-0.088 (0/5)	-0.088 (2/5)	-0.082 (0/5)
1%	-0.114 (0/5)	-0.138 (1/5)	-0.083 (0/5)
2%	-0.091 (0/5)	-0.150 (1/5)	-0.040 (0/5)
5%	-0.121 (0/5)	-0.141 (1/5)	-0.011 (1/5)
10%	-0.134 (0/5)	-0.133 (0/5)	-0.000 (2/5)
20%	-0.145 (0/5)	-0.126 (0/5)	-0.002 (2/5)

E Native SOFisher run

We ran SOFisher from the authors' repository on its bundled mouse cortex data for 5 million steps (seed 222), using the repository's cortex script. The only change was the addition of checkpoint and resume support, so that the run could continue after interruptions. Figure 3 shows the evaluation

reward, which counts captured target regions. Its median over the first 5% of training was -238 and over the last 10% was $+147$ (best 514). Rare evaluation episodes have rewards near -1.7×10^5, so we plot rolling medians.

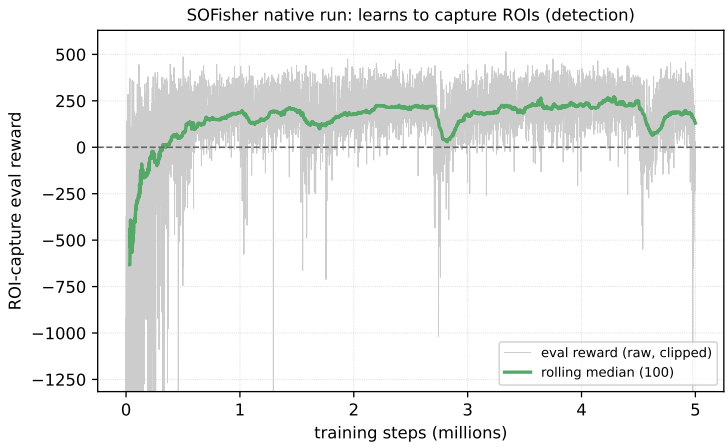


Figure 3: SOFisher evaluation reward during training (rolling median), native run.

F Configuration

All values are fixed in the released `config.py`. Seeds: global 2021; random policy $\{0, \ldots, 4\}$. Imputer (graphflow): image PCA 64, random Fourier features 512 (median-heuristic bandwidth), ridge 10, graph $k = 8$, 20 steps, step size 0.5. k-NN imputer: $k = 10$, joint image and space features. Kriging: image PCA 64, ridge 10, residual $k = 10$. Placement: feature PCA 64, median-heuristic lengthscale, noise 10^{-2}. Error-reduction neighbourhood 15. Value-conditioned: spatial kernel width 0.1 of the slide diagonal, exploration weight 0.5; the ROI-seeking sampler uses the same values. Detection ROIs: top 20% by heterogeneity over 8 spatial neighbours. Statistics: 5,000 bootstrap resamples, exact sign-flip enumeration over study clusters, Holm over 24 primary margins. The code used for the larger slides replaces dense distance computations with sparse and incremental ones; on a 132-run check (11 policies, 3 imputers, 2 slide sizes, 2 budgets) it reproduced the original code's selections and scores exactly.

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